

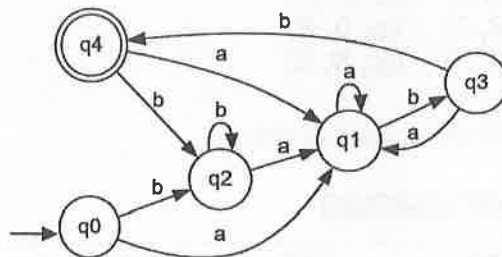
 VIT-AP UNIVERSITY	Final Assessment Test – Short Summer-II (2023-24) - July 2024	
	Maximum Marks: 100	Duration: 3 Hours
Course Code: CSE1008	Course Title: Theory of Computation	
Set No: 3	Exam Type : Closed Book	School: SCOPE
Date: <u>18/07/24</u>	Slot: <u>A</u>	Session: <u>FN</u>
Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice		
General Instructions if any: 1. Assume data wherever necessary. 2. Any assumptions made should be clearly stated.		

PART – A: Answer any TEN Questions, Each Question Carries 10 Marks (10×10=100 Marks)

Q1. Convert the following ϵ -NFA to NFA.

δ	ϵ	0	1	2
$\rightarrow q_0$	q_1	q_0	-	-
q_1	q_2	-	q_1	-
$*q_2$	-	-	-	q_2

Q2. Minimize the following DFA.



Q3. Convert the following regular expression into ϵ -NFA.

$10+(0+11)0^*1$

Q4. For the following DFA, find the equivalent regular expression using Arden's theorem.

δ	0	1
$\rightarrow^*q_1$	q_1	q_2
q_2	q_3	q_2
q_3	q_1	q_2

Q5. Construct Context Free Grammar for the language $L = \{0^{2n}1^n \mid n \geq 1\}$. Using the constructed grammar, write Left Most Derivation of the string 000000111 and draw the corresponding parse tree.

Q6. Check whether the following grammar is ambiguous or not.

**$S \rightarrow iCtS \mid iCtSeS \mid a$
 $C \rightarrow b$**

Q7. Design a PDA that accepts by empty stack for the language $L=\{a^n b^{2n} / n \geq 1\}$ and give Instantaneous Description for the string $w=aabbbb$.

Q8. Convert the following CFG into PDA and give instantaneous description for the string $w=00101011$.

$S \rightarrow 0S1 / A$
 $A \rightarrow 1A0 / S / \epsilon$

Q9. Convert the following CFG into Chomsky Normal Form.

$S \rightarrow a / aA / B$
 $A \rightarrow aBB / \epsilon$
 $B \rightarrow Aa / b$

Q10. Convert the following CFG into Greibach Normal Form.

$S \rightarrow AB$
 $A \rightarrow BS / b$
 $B \rightarrow SA / a$

Q11. Design Turing machine for $L=\{a^n b^n / n \geq 1\}$.

Q12. Consider the following Turing Machine.

State	Symbol		
	0	1	B
q_0	(q_1, B, R)	(q_5, B, R)	—
q_1	$(q_1, 0, R)$	$(q_2, 1, R)$	—
q_2	$(q_3, 1, L)$	$(q_2, 1, R)$	(q_4, B, L)
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	(q_0, B, R)
q_4	$(q_4, 0, L)$	(q_4, B, L)	$(q_6, 0, R)$
q_5	(q_5, B, R)	(q_5, B, R)	(q_6, B, R)
q_6	—	—	—

Give the Instantaneous Description for the following string:

i) 0000100

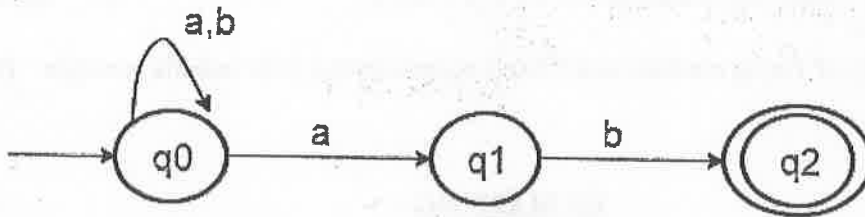
QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1,3	1,2	2	10
Q2	1	1	1,3	1,2	2	10
Q3	2	2	1,2,3	1	2,3	10
Q4	2	2	1,2,3	2	2,3	10
Q5	3	2	1,2,3	2	2,3	10
Q6	3	2	1,2,3	2	2,3	10
Q7	4	2,3	1,2,3,3,4,5	3	1,2,3	10
Q8	4	2,3	1,2,3,3,4,5	3	1,2,3	10
Q9	5	3	3,4,5	2,3	1,2,3	10
Q10	5	3	3,4,5	2,3	1,2,3	10
Q11	6	4	3,4,5	2,3	1,2,3	10
Q12	6	4	3,4,5	2,3	1,2,3	10

 VIT-AP UNIVERSITY	Final Assessment Test – Long Summer (2023-24) - July 2024	
	Maximum Marks: 100	Duration: 3 Hours
Course Code: CSE1008	Course Title: Theory of Computation	
Set : 1	Exam Type : Closed Book	
Date: 18/07/24	Slot: A	Session: AN
Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice		
General Instructions if any: 1. "fx series" - non Programmable calculator are permitted: YES 2. Reference tables permitted : NO		

PART – A: Answer any TEN Questions, Each Question Carries 10 Marks (10×10=100 Marks)

1. Convert the following NFA to its equivalent DFA. (10 M)



2. Citing an example, Explain the model, 5-tuple representation and transition diagram representation of Finite Automata. (10 M)
3. Generate the regular expression that generates all strings starting with 00 or 11 over an alphabet {0,1}. Also, convert the regular expression to its equivalent NFA. (10 M)
4. State pumping lemma for regular languages. Prove that the following language $L = \{ a^n b^n \mid n \geq 1 \}$ is not a regular language. (10 M)
5. What is an ambiguous grammar? Show that the following grammar is ambiguous, where E is the start symbol.

$$E \rightarrow E+E \mid E-E \mid E * E \mid E/E \mid (E) \mid a$$
(10 M)
6. Explain the following with suitable example (10 M)
 - i. Context Free Grammar
 - ii. Regular Grammar
 - iii. Left Linear Grammar
 - iv. Right Linear Grammar
7. Construct PDA for the language $L = \{ a^m b^n c^n d^m \mid m, n \geq 1 \}$ (10 M)

8. Construct the CFG for the PDA $M = (\{q_0, q_1\}, \{0, 1\}, \{R, Z_0\}, \delta, q_0, Z_0, \Phi)$ and δ is given by
- $$\begin{aligned}\delta(q_0, 1, Z_0) &= (q_0, RZ_0) \\ \delta(q_0, 1, R) &= (q_0, RR) \\ \delta(q_0, 0, R) &= (q_1, R) \\ \delta(q_1, 0, Z_0) &= (q_0, Z_0) \\ \delta(q_0, \epsilon, Z_0) &= (q_0, \epsilon) \\ \delta(q_1, 1, R) &= (q_1, \epsilon)\end{aligned}$$
- (10 M)
9. When a grammar is said to be in CNF? Convert the following CFG to CNF.
- $$\begin{aligned}S &\rightarrow ASB \mid \epsilon \\ A &\rightarrow aAS \mid a \\ B &\rightarrow SbS \mid A \mid bb\end{aligned}$$
- (10 M)
10. Explain various closure and decision properties of CFLs. (10 M)
11. Design a TM to recognize the language $L = \{ 0^n 1^n 0^n \mid n \geq 1 \}$ (10 M)
12. Explain the working of Turing machine and 7-tuple representation with suitable example. (10 M)

QP MAPPING

Q. No.	E/A/T	Module Number	Marks	BL	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped
Q1	E	1	10	3	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--
Q2	A	1	10	4	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--
Q3	E	2	10	3	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--
Q4	E	2	10	4	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--
Q5	A	3	10	4	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--
Q6	A	3	10	3	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--
Q7	E	4	10	3	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--
Q8	A	4	10	3	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--
Q9	T	5	10	4	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--
Q10	A	5	10	2	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--
Q11	T	6	10	3	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--
Q12	A	6	10	3	1,2,3,4	1,2,3,4,5	1,2,3,4,5	--

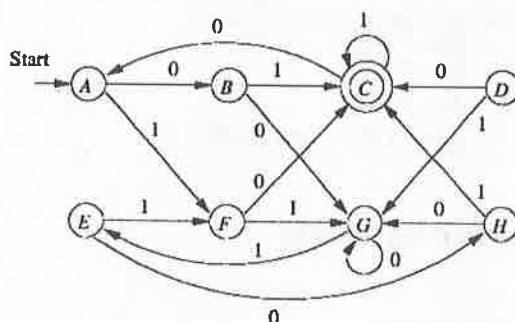
 VIT-AP UNIVERSITY	Final Assessment Test – Short Summer-II (2023-24) - July 2024	
	Maximum Marks: 100	Duration: 3 Hours
Course Code: CSE1008	Course Title: Theory of Computation	
Set No: 2	Exam Type : Closed Book	School: SCOPE
Date: 20/07/2024	Slot: C	Session: FN
Keeping mobile phone/smart watch, even in 'off' position is treated as exam malpractice		
General Instructions if any: 1. Assume data wherever necessary. 2. Any assumptions made should be clearly stated.		

PART – A: Answer any TEN Questions, Each Question Carries 10 Marks (10×10=100 Marks)

Q1. Design a NFA for the language $L = \{ a(a+b)^*b \}$ and validate the following strings.

$w_1 = \text{aaaab}$; $w_2 = \text{abbba}$

Q2. Minimize the following DFA.



Q3. Convert the following regular expression into ϵ -NFA.

$(a+b)a^*(b+a)^*b$

Q4. Find the regular expression for the given DFA using Arden's theorem.

δ	a	b
$\rightarrow q_0$	q_1	q_1
q_1	q_2	q_2
$*q_2$	q_2	q_2

Q5. Write the context-free grammar for the language $L = \{ a^i b^{2i} \mid i \geq 1 \}$ and derive the following strings using left most derivation.

$w_1 = \text{aabbbb}$; $w_2 = \text{aaabbbbbbb}$

Q6. Show that the following grammar is ambiguous.

$E \rightarrow E + E$
 $E \rightarrow E * E \mid id$

Q7. Design a PDA that accepting by final state for the language $L = \{ a^n b^m c^n \mid n, m \geq 1 \}$ and give Instantaneous Description for the string $w = \text{aacbb}$.

Q8. Convert the following CFG into PDA and perform instantaneous description for the string $w=0011$.

$S \rightarrow A1B$
 $A \rightarrow 0A \mid \epsilon$
 $B \rightarrow 0B \mid 1B \mid \epsilon$

Q9. Convert the following CFG into Chomsky Normal Form.

$S \rightarrow ASB \mid \epsilon$
 $A \rightarrow aAS \mid a$
 $B \rightarrow SbS \mid A \mid bb$

Q10. Convert the following CFG into Greibach Normal Form.

$S \rightarrow AA \mid 0$
 $A \rightarrow SS \mid 1$

Q11. The function $f(n)=n+1$ is computable. Design a Turing machine for the function accepting the strings over $\Sigma = \{0\}$.

Sample Input String: 000 **Sample Output String:** 0000
 Give Instantaneous description for the input string 0000.

Q12. Consider the following Turing Machine.

State	Symbol				
	0	1	X	Y	B
q_0	(q_1, X, R)	—	—	(q_3, Y, R)	—
q_1	$(q_1, 0, R)$	(q_2, Y, L)	—	(q_1, Y, R)	—
q_2	$(q_2, 0, L)$	—	(q_0, X, R)	(q_2, Y, L)	—
q_3	—	—	—	(q_3, Y, R)	(q_4, B, R)
q_4	—	—	—	—	—

Show the ID's reachable from the initial state if the input is:

- i) 0011
 ii) 0001

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	1	1,3	1,2	2	10
Q2	1	1	1,3	1,2	2	10
Q3	2	2	1,2,3	1	2,3	10
Q4	2	2	1,2,3	2	2,3	10
Q5	3	2	1,2,3	2	2,3	10
Q6	3	2	1,2,3	2	2,3	10
Q7	4	2,3	1,2,3,3,4,5	3	1,2,3	10
Q8	4	2,3	1,2,3,3,4,5	3	1,2,3	10
Q9	5	3	3,4,5	2,3	1,2,3	10
Q10	5	3	3,4,5	2,3	1,2,3	10
Q11	6	4	3,4,5	2,3	1,2,3	10
Q12	6	4	3,4,5	2,3	1,2,3	10